# Ayush Kumar

+91-7032853030 | theofficialayush.kumar@gmail.com | AyushK0808 | ayushk0808 | ayushk08.com

Software engineer working at the intersection of **ML systems and backend infrastructure** — building LLM agents, document intelligence, and distributed pipelines in .NET, Go, Rust, and Python. Peer-reviewed research (**ICACKE 2026**); first-place international robotics finish (**ISDC 2024**).

## Experience

**Cornerstone OnDemand** Dec 2025 – Present
*Junior Engineer* *Mumbai, India*

- Enhanced the **CSX Scheduler**, an enterprise distributed job-orchestration platform on **.NET** — REST APIs, job lifecycle & retry mechanisms, and **React** operational dashboards for workloads across **AWS EC2** environments
- Built a mass PDF-generation pipeline for HR onboarding forms: a **Python** data/HTML transformation engine feeding a **Rust** parallel rendering service, deployed in **22+ production regions**, generating thousands of PDFs per batch
- Developed **CSOD Harness**, an internal LLM engineering assistant with Roslyn source-code indexing, a code–infrastructure knowledge graph, and multi-model agent orchestration, cutting token consumption by **56%** in multi-repo incident debugging

**Deloitte USI** Jun 2025 – Aug 2025
*SDE Intern — AI & Backend Engineering* *Hyderabad, India*

- Developed backend services for an AI-enabled internal productivity tool using **FastAPI**, building RESTful APIs within a microservice architecture
- Containerized services with **Docker**; achieved **90%+** PyTest coverage, ensuring stability across deployments

**Samsung R&D Institute India** Sep 2024 – Jun 2025
*Research Intern — ML & Generative AI* *Remote*

- Built an end-to-end customer segmentation pipeline (Pandas, Scikit-learn, Lifetimes) with **RFM modeling** and **CLV estimation** (BG/NBD, Gamma-Gamma) on enterprise datasets
- Designed K-Means/K-Medoids segmentation and a **GenAI workflow** converting segment outputs into executable SQL, automating audience selection and campaign targeting

**Titan Smart Labs** Jun 2024 – Aug 2024
*Algorithm Intern — Smart Wearables* *Hyderabad, India*

- Built preprocessing pipelines for multimodal smartwatch time series (heart rate + accelerometer): filtering, synchronization, imputation, and feature engineering
- Improved sleep & stress classification using Scikit-learn models with leave-one-subject-out CV, hyperparameter tuning, and ROC/PR benchmarking

## Leadership

**Technical Head** — *VinnovateIT*

- Deployed production apps on **Cloudflare & AWS** serving **30k+ campus users**; led **15+ full-stack projects**
- Designed software systems for hostel operations, including applications for operational coordination and student services
- Mentored **200+ members** in AI/ML and web dev; organized **VinHack**, a national hackathon with **300+ participants**

**Software Lead** — *Team Ardra (Autonomous Drones)*

- **Won ISDC 2024** (International Space Drone Challenge), leading the team's autonomous software stack to first place
- Built the vision (**YOLO, SegFormer, SAM**) + **SLAM** pipeline with **ROS** and edge inference on Raspberry Pi & Jetson Nano

## Education

**Vellore Institute of Technology** 2022 – 2026
*B.Tech, Computer Science & Engineering* *GPA: 9.35/10*

## Projects

**Sentiment-Augmented Financial Advisor**
*Python, Streamlit, LangChain, Ollama (LLaMA 3.2), SQLite*

- AI investment-analysis system fusing market indicators, investor profiling, and news sentiment into explainable buy/hold/sell signals — **accepted at ICACKE 2026**
- Privacy-preserving **local LLM inference** with RAG context aggregation; modular dashboard for portfolio analytics and natural-language queries

**MailForge — Mass-Mailing Platform**
*Go, Next.js, PostgreSQL, Redis, Docker, NGINX*

- Distributed, self-hosted email-campaign platform: **Go** API, **Redis-queued worker pool**, and Next.js dashboard for contacts, templates, and campaigns
- JWT-scoped multi-tenancy and rate-limited concurrent delivery (**20 goroutines, 100 sends/sec**) with retry-backoff; fully Dockerized with GitHub Actions CI

**Multi-Agent Form Schema ETL Pipeline**
*Python, LayoutLMv3, Donut, LoRA/PEFT, LangGraph, Groq*

- Layout-aware document extraction running **LayoutLMv3 + Donut in parallel**, fused via a confidence-weighted reflexive policy with cross-field consistency validation
- **LangGraph 8-node pipeline** with LoRA-adapter routing, Groq-based schema normalization over a MiniLM embedding registry, and schema-driven SQLite population

## Technical Skills

**Languages:** Python, Go, Rust, Java, JavaScript, TypeScript, C, C++, SQL, R
**Frameworks:** React, Next.js, Node.js, FastAPI, Flask, .NET, TensorFlow, Keras, LangChain, LangGraph, Streamlit
**Tools & Platforms:** Docker, Redis, PostgreSQL, MongoDB, MySQL, Jenkins, Git, AWS, Cloudflare, Linux